

Samantha Ajovalasit

Profile

Data Scientist with a PhD in Economics, Management and Statistics, specialised in NLP, Large Language Models, and quantitative risk modelling. Development of end-to-end text pipelines (extraction, classification, fine-tuning) using spaCy and HuggingFace Transformers; LLM deployment via Ollama (LLaMA, Mistral, Deepseek). Full-stack decision-support apps shipped to cloud production (WhatDSApp/AMELIA, EUMEPLAT/shinyapps.io). Research published at Bruegel (EU think tank) and in peer-reviewed journals.

Experiences

2023 – 2026 **Researcher RTD (a), University of Palermo**

Project *GRINS – Growing Resilient, Inclusive and Sustainable*. Development of pipelines in multiple environments for the analysis of complex socioeconomic data; statistical modelling and machine learning for decision support on public policies. Development of an interactive web-app (WhatDSApp) integrated into the AMELIA cloud platform. Tenure track in Mathematics for Business.

2022 – 2023 **Research Fellow, University of Palermo**

Project *Just Smart – “Smart Justice: Tools and Models to Optimise Judges’ Work”*. Development of algorithms to evaluate and improve court efficiency. Analysis of unstructured court data; NLP analysis on Italian judicial documents: data extraction, text classification and sentiment analysis to identify bottlenecks and propose data-driven solutions.

2021 – 2022 **Research Fellow, Ca’ Foscari University, Venice**

Project *H2020 Grant 101004488 European Media Platform – EUMEPLAT*. Development of analysis algorithms to explore the spread of anti-European sentiment in unstructured data.

Contract Lecturer – Supplementary module *Advanced Application of Computer Science for Cultural Heritage* – Ca’Foscari University Venice, basics of R and network analysis with interactive training and coding.

2017 **Internship, RJC Soft s.r.l., Pisa**

Project *Sprint*. I worked with the R&D team, where I analyzed, using Python and QGIS, the urban traffic system in the municipality of Pisa.

Education and Training

2018 – 2020 **PhD in Economics, Management and Statistics, University of Messina, XXXIII Cycle, EQF 8**

Thesis: “Disorientation towards routine immunization, COVID-19 anxiety, and propensity to vaccinate. Analysis based on Twitter data in Italy”. Defended: 21 December 2020.

Large-scale NLP analysis of Italian Twitter data: sentiment analysis, text classification and model fine-tuning for detecting vaccination hesitancy and measuring the impact of political communication on vaccination propensity.

2017 – 2018 **II Level Master’s in Big Data Analytics & Social Mining, University of Pisa, EQF 7**

Thesis: *Traffic Analysis in Urban Environments via GPS Trajectories*. Defended: 4 May 2018. Covers data mining, machine learning, data analysis & visualization, network science, computational social science, ethics, data journalism. Focus on big data management, knowledge extraction, and decision support - (Python, R, SQL, Knime).

- 2013 – 2016 **Master's Degree in Economics**, *University of Pisa, Scuola Superiore Sant'Anna*, EQF 7
Advanced economics training with quantitative and statistical tools. Thesis: *Option Pricing with Monte Carlo Simulation and Variance Reduction Technique*. Defended: 5 July 2016.
- 2007 – 2012 **Bachelor's Degree in Economics and Finance**, *University of Palermo*, EQF 6
Thesis: *"Loan Transfer and Securitisation"*, Defended: 5 November 2012.

Additional Training

- 2022 **Summer School in Network Econometrics**, *Italian Society of Econometrics - Ca' Foscari University of Venezia*
The course aims to provide the fundamentals of the econometrics network with particular reference to the Network mapping and visualisation, the Network Extraction Methods, Multi-layer Network Models, and their applications to finance.
- 2019 **Summer School in Data Science and Epidemic Models**, *University of Trento*
Complex data analysis, digital tracking, human mobility, computational methods and social media analytics.
- 2019 **Fundamental of Digital and Computational Demography**, *Max Planck Institute for Demographic Research*, Rostock
The course provides tools to improve demographic analysis in demography and social science methods, as well as to interpret new kinds of data, focusing in particular on big data.
- 2018 **Summer School on Clustering, Data Analysis and Visualization of Complex Data**, *University of Catania*
Clustering, classification, visualisation of large datasets; numerical analysis and cloud-computing platforms.

Teaching Activities

- 2023 – 2026 **Lecturer for the course *Mathematics for Business***, *Bachelor's Degree in Economics and Business Administration – Class L-18 – University of Palermo.*, Role: Course holder.
- 2018 – 2020 **Senior Tutor**, *University of Catania*
Course: *Data Science and Demographic Processes*. Module dedicated to NLP, sentiment analysis and REST API integration for social media data analysis – Twitter.
- 2018 – 2020 **Senior Tutor**, *University of Catania*
Course: *Economic Statistics*, Bachelor's Degree in Business Administration.

Software and Tools Released

- WhatDSApp Interactive web-app for public debt assessment. Integrates sustainability analysis to create optimal financing scenarios and strategies. AMELIA platform, GRINS project: <https://ameliadp.grins.it/>
- Dashboard Interactive dashboard for the EUMEPLAT project – Data clustering reports. <https://eumeplat.shinyapps.io/EumeplatDashboard/>

Technical Skills

LLM & Generative AI	Deployment and management of Large Language Models via Ollama with Open WebUI interface. Integration of open-source models (LLaMA, Mistral, Gemma, Deepseek) into analytical workflows. Supervised model fine-tuning on text corpora (Twitter, judicial documents, European media). Prompt engineering, output evaluation and LLM integration into production pipelines. Claude Code (CLI) for AI-assisted development in agentic environments.
NLP & ML	spaCy, scikit-learn, NLTK, Transformers, Gensim, Word2Vec/Doc2Vec for text processing (data extraction, classification, topic recognition, clustering, sentiment analysis, opinion mining). Annotation pipeline: Doccano, spaCy DocBin, spaCy CLI (training from scratch, <code>it_core_news_lg</code>). Italian judicial NLP: NER on legal corpus (displacy). Sentiment scoring: VADER, SentiWordNet. Feature extraction: TF-IDF, CountVectorizer. Data collection: Selenium, REST API, Tweepy.
Big Data & Data Engineering	Apache Hadoop (MapReduce), Spark. NoSQL: MongoDB. Relational and spatial databases: SQL/MySQL. ETL pipelines. End-to-end pipeline architecture: data ingestion, transformation, analysis, output.
Web App & Data Visualisation	Full-stack web application built from scratch: Bootstrap 4, HTML/CSS, Plotly/Dash, Shiny. Input validation, real-time status display. Financial outputs: fan charts, efficient frontier, endogenous/exogenous variable charts. Public dashboards.
Quantitative Finance & Stochastic Modelling	(Pseudo) Monte Carlo simulation with variance reduction techniques. Stochastic Debt Sustainability Analysis (SDSA). ESG risk factor modelling, scenario analysis, sensitivity analysis. Portfolio optimisation (efficient frontier, CVaR). Option Pricing. Financial Mathematics teaching. Financial data APIs: Refinitiv Eikon (bond data, yield curves, credit spreads). Network econometrics.
Languages & Tools	Python (advanced), R, MATLAB, Stata, GAMS, Gretl, Knime. REST APIs, JSON, Docker, Git/GitHub. Environments: VS Code, Jupyter, $\text{\LaTeX}$, LibreOffice. Network analysis and visualisation (R, Python). QGIS.
Systems & Infrastructure	Linux (Arch, Ubuntu), macOS, Windows. Workstation administration, shell scripting. Remote development via VS Code Remote SSH on Linux machines. Cloud platforms (AMELIA, GRINS). Sysadmin experience in computational research environments.
Languages	Italian (native), English (B2 – upper intermediate), French (B1 – intermediate).

Proceedings, Papers and Publications

- 2025 **Are bad governments threat to sovereign defaults? Effects of political risk on debt sustainability.** Working Paper 01/2025, Bruegel. [Bruegel.org](https://bruegel.org)
- 2025 **How much of threat to US debt sustainability is Trump's One Big Beautiful Bill Act?** Analysis Report. Bruegel. [Bruegel.org](https://bruegel.org)
- 2025 **ESG factors and sovereign debt.** *Economia Italiana* 46(3), 145-164. DOI: 10.57621/EI2024-03-I-06
- 2024 **Debt Sustainability in Context of Population Ageing: Risk Management Approach.** *Risks* 12(12). <https://www.mdpi.com/2227-9091/12/12/188>
- 2022 **H2020 European Media Platform - EUMEPLAT - Data clustering reports** - https://www.eumeplat.eu/wp-content/uploads/2022/03/D1.5_Data-Clustering-Report.pdf

2021 **Evidence of disorientation towards immunization on online social media.** PlosOne.
<https://doi.org/10.1371/journal.pone.0253569>

Scientific Society Memberships

AMASES, Italian Association for Mathematics Applied to Social and Economic Sciences.
Member since 2023.

Member of Scientific or Organizing Committee

Member of Organizing Committee - XXVI Quantitative Finance Workshop - University of Palermo - The workshop aims to bring together academia and industry to stimulate discussions and interactions between researchers and professionals on the theoretical and applied aspects of Quantitative Finance broadly understood. 15-17 April 2025

Member of Scientific Committee - The international Conference on Frontiers in Stochastic Modelling for Finance - University of Palermo - This conference aims to enhance existing connections and foster new collaboration between researchers in the fields of modeling for finance, stochastic control, and quantitative finance. 25-27 October 2023

Member of Scientific Committee - X Conference of International Association for Tourism Economics - IATE - University of Palermo, 23-26 June 2026 (upcoming)